

Sentiment Analysis of Product Reviews

Turning Opinions into Insights



Team Members



Sumangala Gouda
4SF23CS220



Sushmita S
4SF24CS421



Sanjana KS
4SF23CS192

*Reviews
Speak,
We Understand!*

Introduction

- Online platforms generate a **huge volume** of customer reviews about products and services.
- These reviews contain **valuable information** about customer opinions, satisfaction, experiences and preferences.
- Manually analyzing a large number of reviews is **time-consuming, subjective** and **difficult**.
- Sentiment analysis, a **Natural Language Processing (NLP)** task, helps in automatically identifying the sentiment expressed in text.

“ Every review has a story. Let AI understand it. ”



Problem Statement

- “To automatically analyze textual reviews and classify them into **Positive or Negative sentiment** using a **deep-learning-based NLP model**.”



*From opinions
to insights*

01



Study Sentiment Analysis

To understand and analyze sentiment from textual product reviews.

02



Preprocess Review Text

To clean and preprocess raw review text and convert it into a suitable numerical representation.

03



Develop a BiLSTM Model

To build a deep learning-based sentiment classification framework using BiLSTM.

04



Classify Sentiments

To classify product reviews into Positive or Negative sentiment categories.

05



Provide Confidence & Insights

To display prediction confidence and useful statistical information through visualizations.

06



Demonstrate via Web App

To implement and demonstrate the complete pipeline through an interactive Streamlit application.

Turning
Reviews into
Insights



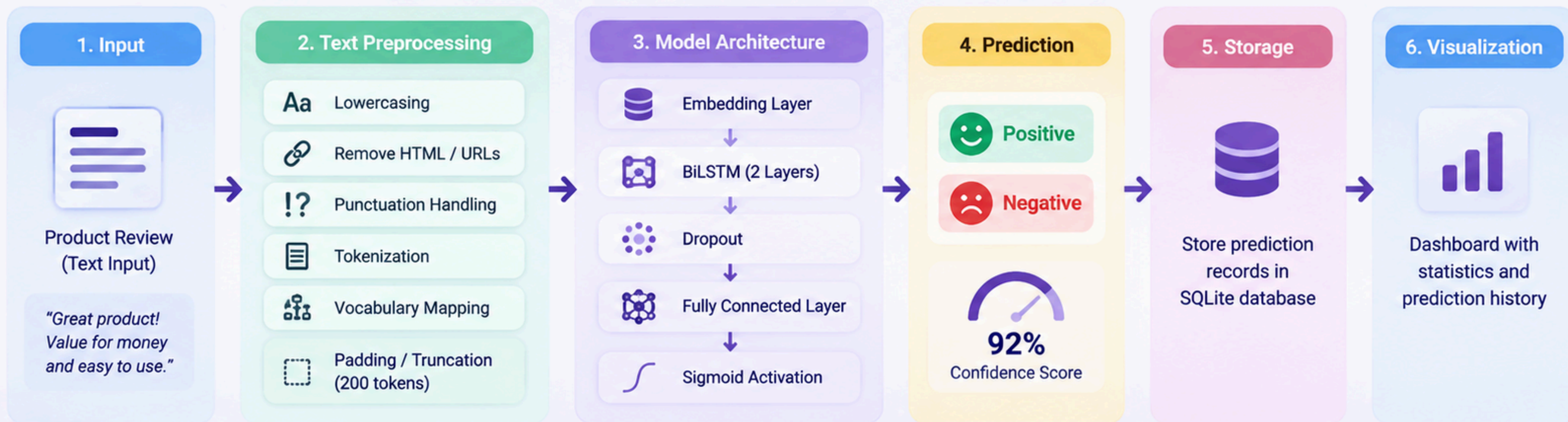
“ Our goal is to make sense of every review. ”



Overview

The proposed system takes a product review as input, preprocesses the text, converts it into a numerical representation and uses a Bidirectional LSTM model to predict the sentiment as Positive or Negative. The application also provides prediction confidence and stores the results for visualization.

*A complete pipeline
from text to insight*



End-to-end pipeline: Preprocess → Classify → Visualize → Gain Insights



Technology Stack

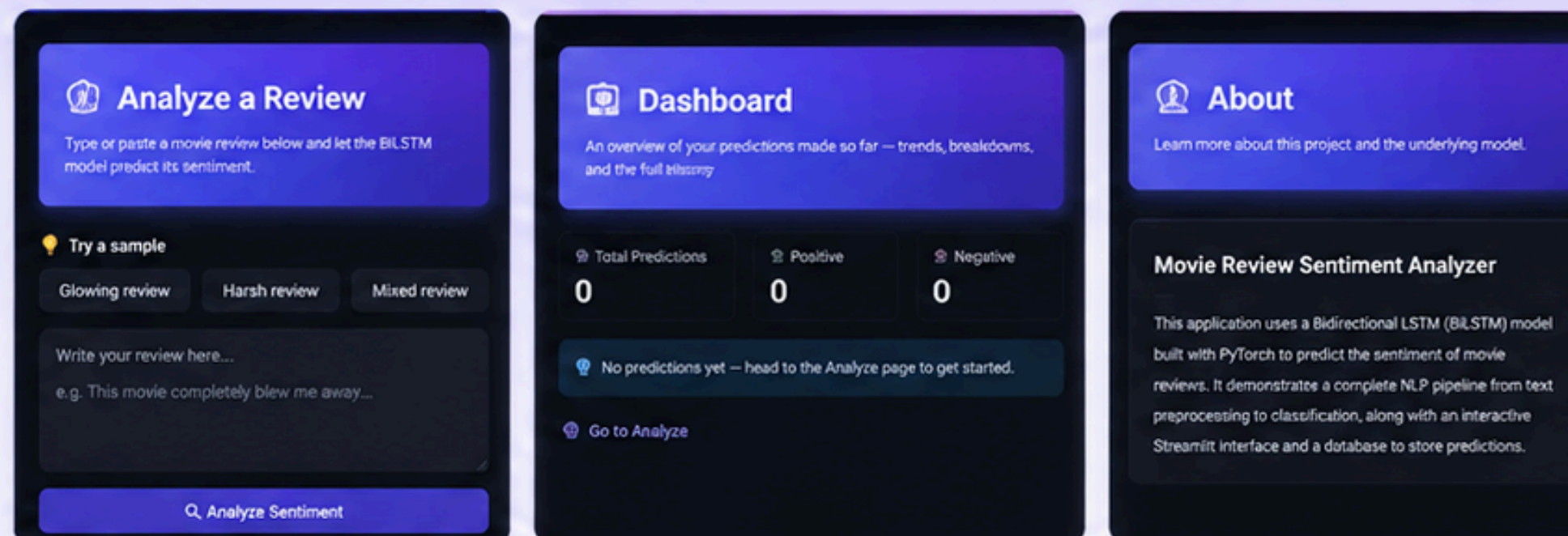
Key Technologies Used

Technology	Purpose
 Python	Application development
 PyTorch	Deep learning model and inference
 Streamlit	Interactive web interface
 SQLite	Prediction storage
 Pandas	Data processing
 Altair	Visualization



Application Interface

An interactive Streamlit application



1 Analyze Page

- Enter a review (movie/product)
- Analyze sentiment
- View prediction result
- See confidence score
- Try sample reviews

2 Dashboard Page

- View total predictions
- Positive/negative distribution
- Prediction history (sorted by date)
- Search and filter records
- Export data (CSV)
- Visual insights and charts

3 About Page

- Project overview
- Model details
- Technologies used
- How it works
- Future scope



Simple & User-friendly

Built with Streamlit for easy access



End-to-End Solution

From input to insights



Scalable Framework

Applicable to product reviews

*Technology turns
opinions into opportunities*



Application Output

Analyze Page

Analyze a Review

Type or paste a movie review below and let the BiLSTM model predict its sentiment.

Try a sample

Glowing review

Harsh review

Mixed review

Write your review here...

The movie was absolutely fantastic! The acting was brilliant and the storyline kept me engaged throughout.

Analyze Sentiment

User-friendly interface to analyze review sentiment



Sample Predictions

Example results from the application

"An amazing movie with great performances. Highly recommend!"



Positive

Confidence: 0.94

"The plot was dull and the acting was disappointing."



Negative

Confidence: 0.87

"Good visuals and music, but the story could have been better."



Positive

Confidence: 0.76

The model provides sentiment prediction with confidence score for each review.



Dashboard Insights

Visualization of predictions



Dashboard showing sentiment distribution and prediction history.



Model Performance

As reported in the project description

~83%

Accuracy (Test Dataset)

- Binary sentiment classification (Positive / Negative)
- Confidence score for each prediction
- Results based on original experiments (training/evaluation pipeline not included in repository)



Key Findings

What we observed from the prototype

- BiLSTM model effectively classifies review sentiments.
- Preprocessing improves model performance.
- Streamlit enables an interactive and user-friendly interface.
- Confidence scores help users understand prediction context.
- SQLite database successfully stores prediction records.
- Dashboard provides useful visual insights.

Meaningful
insights from
every review



Limitations

Current challenges in the implementation



Binary Classification Only

The model classifies reviews only as Positive or Negative. Neutral or mixed sentiments are not handled.



Mixed Reviews

Reviews containing both positive and negative aspects may be misclassified.



Input Length Limitation

The model uses a maximum sequence length of 200 tokens. Very long reviews may lose important context.



Confidence Interpretation

The prediction confidence score is not equivalent to a statistically validated confidence interval.



Training & Evaluation Pipeline

The complete training and evaluation pipeline, datasets, and detailed metrics are not included in the repository.



Future Scope

Opportunities for enhancement and expansion

01



Neutral & Mixed Sentiment Classification

Extend the model to handle neutral and mixed sentiments for more fine-grained analysis.

02



Aspect-based Sentiment Analysis

Identify sentiment towards specific product aspects (e.g., quality, price, design, service).

03



Transformer-based Models

Explore advanced models such as BERT, RoBERTa or other transformer architectures for improved performance.

04



Multilingual Sentiment Analysis

Extend the framework to support multiple languages for a global user base.

05



Cloud Deployment & Scalable Database

Deploy the application on cloud platforms and use a centralized database for large-scale real-time analysis.

More
Insights,
A Brighter
Tomorrow.



Explore
Improve
Expand
Create Impact



Final Takeaway

While the current prototype has certain limitations, it provides a strong foundation for advanced sentiment analysis systems with real-world applicability in product reviews.



Overall Conclusion

The project demonstrates an end-to-end sentiment analysis pipeline using Natural Language Processing, BiLSTM, PyTorch and Streamlit. It effectively processes review text, performs sentiment prediction, provides confidence information, stores prediction records and visualizes results through an interactive dashboard.

Although the current implementation is demonstrated on movie reviews, the underlying framework is applicable to product-review analysis and can be extended for real-world use cases.

“Turning feedback into meaningful insights”



Key Takeaways



Deep learning models like BiLSTM are effective for sentiment analysis.



Streamlit enables an interactive and user-friendly interface.



Confidence scores and visualizations help users understand review sentiment.



The framework can be adapted to product reviews and other domains.



Key Impact

- ✓ Automates review analysis
- ✓ Helps businesses understand customer opinions
- ✓ Provides quick and visual insights
- ✓ Can be extended for real-world applications



Final Thought

Sentiment analysis bridges the gap between customer opinions and data-driven decisions, helping create better products and a better user experience.